Syed Zuhair Hossain

Khilgaon, Dhaka, Bangladesh syed.zuhair.hossain@g.bracu.ac.bd +88-01756843882
syed-zuhair-hossain ZuhairHossain

Career Objective

With a profound passion for Human-Computer Interaction and a commitment to social good, I aspire to develop transformative, low-cost technologies that address the needs of low-literate populations in Bangladesh. My extensive field research experience provides invaluable insights into these communities, driving my dedication to bridge the digital divide. Eager to collaborate with visionary scholars and institutions, I aim to harness technology to foster inclusivity, enhance accessibility, and create a more equitable society.

Education

BSc **Brac University**, Computer Science & Engineering — Dhaka, Bangladesh
Jan. 2019 to May 2023

Notable Coursework: Human Computer Interface, Neural Networks, Artificial Intelligence, Software Engineering, Introduction to Robotics.

HSC **Dhaka City College**, Science Department — Dhaka, Bangladesh
Jan. 2016 to Dec. 2018

- GPA: 5.00/5.00

SSC **Ideal School and College**, Science Department — Dhaka, Bangladesh
Jan. 2010 to Dec. 2016

- GPA: 5.00/5.00

Paper Publications

In review: Oct 2026
Decoding Mental Friction: Understanding Social and Non-social Biases in LLMs Toward Autistic Users Autism in Adulthood 2026

Syed Zuhair Hossain, Valerie De Ryan, Katelyn Pryor, Shameem Ahmed

Examining Large Language Models Within Autism-Related Contexts: A Systematic Review of Bias and (Mis)Representation July 2025
COMPSAC 2025
Read Paper

Katelyn Pryor, Thorleif Coleman, ***Syed Zuhair Hossain***, Eric Tootill, Shameem Ahmed,

COMPSAC 2025: IEEE International Conference on Computers, Software, & Applications

MonChitro: Unveiling The Stress and Depression About Future Among Undergraduate Students Dec. 2024
NSysS'24
Read paper

Syed Zuhair Hossain, Anika Tahsin Miami, Jannatun Noor Mukta

NSysS'24: 11th International Conference on Networking, Systems and Security 2024

Poster Publications

NavConnect: Empowering Boatmen Through USSD-based Transportation Services Feb. 2024
HOTMOBILE '24
Read Poster Paper

Syed Zuhair Hossain, Anika Tahsin Miami

HOTMOBILE '24: Proceedings of the 25th International Workshop on Mobile Computing Systems and Applications

Psychological Aspects and Potential Screening of Depression Among Cancer Patients Dec. 2023
NSysS'23
Read Poster

Anika Tahsin Miami, Anika Priodorshinee Mrittika, ***Syed Zuhair Hossain***, Jannatun Noor Mukta

NSysS'23: 10th International Conference on Networking, Systems and Security 2023

Undergrad Thesis

Beyond words: an exploration of free-hand sketches and EEG-based neurobiological signatures to unveil the underlying depression among cancer patients
Anika Tahsin Miami, Anika Priodorshinee Mrittika, ***Syed Zuhair Hossain***, Jannatun Noor Mukta

April 2023
Brac University
Read Paper

Research Grant

Research Seed Grant Initiative 2023
Analyzing Psychological As Well As Physiological Aspects of Cancer Patients And Dealing with Their Griefs through Computing Solutions
Anika Tahsin Miami, Anika Priodorshinee Mrittika, ***Syed Zuhair Hossain***, Jannatun Noor Mukta

July 2023
Brac University

Research Work Experience

Empathetic Computing Lab, *Virtual Internship*
- The project just kicked off with the aim of researching EEG-based user input preference detection.

University of South Australia & University of Auckland
Sept. 2024 to Apr. 2024

NEAT (Next Generation Effective Affective Technology),
Remote Research Collaboration
- Mental health analysis of neurodivergent individuals, focusing on identifying patterns and developing strategies for improved well-being.

Western Washington University, USA
June. 2024 to **Present**

NSUHCI DIAL, *Undergrad Data Science Research Assistant*
- Alor Akash: Intersection of gender and technology in Bangladesh
- Japan Mental Health Dataset in Corona Pandemic

Dhaka, Bangladesh
Oct. 2021 to Feb. 2022
5 months

Professional Work Experience

Brac University, Instructor, App Development
- Writing clean, maintainable, and efficient Dart code for Flutter framework.
- Collaborating with designers to implement UI/UX designs.
- Implementing features and functionality as defined by the project requirements.
- Assisting in the creation of user guides or help files for the end-users.
- Writing unit and integration tests to ensure code quality and reliability.

Savar, Dhaka, Bangladesh
Sept. 2022 to **Present**

Therap BD Ltd., DevOps Engineer
- Building Docker images for applications and services using Jenkins pipeline.
- Deploying and managing containerized applications using Kubernetes.
- Using Ansible for configuration management and orchestration of server environments.

Dhaka, Bangladesh
Sept. 2025 to **Present**

RedDot Digital IT Ltd., Assistant DevOps Engineer
- Provisioning and managing cloud resources such as EC2 instances, RDS databases, and S3 buckets on AWS.
- Building Docker images for applications and services.
- Deploying and managing containerized applications using Kubernetes.
- Using Ansible for configuration management and orchestration of server environments.

Dhaka, Bangladesh
Feb. 2024 to Sept. 2025
1 year and 7 months

RedDot Digital IT Ltd., Trainee DevOps Engineer
- Automating deployment pipelines and workflows using GitLab CI/CD.

Dhaka, Bangladesh
June 2023 to Jan. 2024

- Writing and maintaining shell scripts for automating tasks and managing systems.
- Monitoring system alerts and logs for issues using Prometheus & Grafana as well as Elasticsearch stack.
- Configuring Grafana dashboards for visualizing system and application performance.

7 months

RedDot Digital IT Ltd., DevOps Engineer, Intern

Dhaka, Bangladesh
Feb. 2023 to May 2023
7 months

- Setting up and maintaining monitoring solutions using Prometheus for metrics collection.
- Configuring Grafana dashboards for visualizing system and application performance.
- Integrating Elasticsearch for log aggregation and analysis.
- Monitoring system alerts and logs for issues.

IEEE BracU Student Branch, Webmaster

Dhaka, Bangladesh
Jan. 2021 to Dec. 2022
2 years

- Website Maintenance:
  - Perform regular updates to content, themes, and plugins.
  - Fix bugs and troubleshoot website issues.
- Security & Performance Management:
  - Implement security measures to protect the website from threats.
  - Monitor website performance metrics (load times, uptime) & Optimize images, scripts, and other elements to improve performance.
  - Regularly update software to patch vulnerabilities.

F2p, Web Developer and Designer

Remote, Bangladesh
March. 2021 to Feb. 2022
1 year and 1 month

- Created a couple of WordPress websites for customers
- Content created for customers

Centre for Emotional Intelligence & Innovation - CEII, Intern

Dhaka, Bangladesh
Sep. 2019 to Mar. 2020
7 month

- Administrative works
- Content creation for Promotion
- R&D for student activities and make content for students

Skills

Life skill: Leadership, Photography, Cycling, Graphics Designing, App Development

Automation Tools: Shell Scripting, Docker, Ansible, Terraform

Programming Languages: Java, C++, C, Python, HTML, CSS, JavaScript, PHP, SQL, Dart, YAML

Data Analysis Tools : Jupyter Notebook/Google Colab, Kaggle, Microsoft Power BI

Graphics Design Software : Adobe Illustrator, Photoshop, XD, Figma, Premiere Pro, After Effects

Conference Reviewer

Association of Computing Machinery (CHI'24 & CHI'25): Paper, Case Study and Alt.chi

ACM Conversational User Interfaces (CUI'24): Full Research Paper

ACM Conference on Designing Interactive Systems (DIS'24): Full Research Paper

International Conference on Soft Computing and its Engineering Applications (icSoftComp2022, icSoftComp2023, icSoftComp2024): Full Research Paper

Conference Volunteer

(MobileHCI) 2024
ACM International Conference on Mobile Human-Computer Interaction Melbourne, Australia

(UbiComp / ISWC) 2024
Melbourne, Australia

(CHI PLAY) 2024
Annual Symposium on Computer-Human Interaction in Play Tampere, Finland

(VIS) 2024
IEEE Visualization Florida, USA

Conference Participation

- **International Conference on Soft Computing and its Engineering Applications** (icSoftComp2022, icSoftComp2023, icSoftComp2024)
- **Conference on Human Factors in Computing Systems** - CHI'23
- **Ubiquitous computing** -UbiComp'22

Projects

A Text-Based Healthcare Chatbot to Support Patients 2023
(Project Demo)

- A text-based healthcare chatbot can address this issue by providing 24/7 access to information and personalized support, thus alleviating patient concerns, reducing the burden on healthcare workers, and improving overall patient satisfaction and outcomes.
- Used Python, NLP, Heroku, Docker, Ngrok, and Rasa framework.

Analyzing Maternal Health in Nigeria using Machine Learning 2023

- This project aims to address Nigeria's high maternal mortality rate by using AI algorithms to predict maternal health risks and analyze data to identify patterns and trends. It will develop a user-friendly web app and dashboard for risk assessments and data visualization, ultimately enhancing maternal health outcomes through targeted interventions and accessible insights.
- Used Python.

BracU Faculty Info 2022

- Developed an electronic classroom where multiple users can view and simultaneously draw on a "chalkboard" with each person's edits synchronized.
- Used Flutter and Dart.

ROBUST 2020 to 2021

- ROBUST is a Podcast website where you can hear audio shows and download audio on the personal device.
- Used HTML, CSS, JS, PHP and SQL.

Improving Food Security and Crop Yield in Kenya Through Machine Learning 2021

- I identified underutilized farming land using satellite imaging, applied Kenyan open-source satellite imagery datasets for crop yield prediction, and developed a weather information sharing system for farmers to enhance farming decisions.
- Used JavaScript and Python.

Developing an AI-driven & Secure Online Marketplace 2021

- This project enhances Bangladesh's e-commerce sector by analyzing online marketplaces to identify technology, trust, and transaction issues. It develops AI-driven

solutions for fraud detection, product recommendations, and marketplace improvements, with findings made open source to benefit regulators, businesses, and consumers.

- Used JavaScript and Python.

Professional Membership

Institute of Electrical and Electronics Engineers (IEEE), Student Member — Jan. 2021 to Present

Association for Computing Machinery (ACM), Student Member — Nov. 2023 to Present

Special Interest Group on Computer–Human Interaction (SIGCHI), Student Member — Nov. 2023 to Present

Professional Workshop

Writing for Interdisciplinary Computing Audiences in COMPASS '25 (Workshop Info) — July 2025

Voluntary Leadership Roles

Society for the Popularization of Science, Volunteer & Organizer — Oct. 2021 to Present

BRAC University ACM Student Chapter, Mentor — Sept. 2021 to May. 2023

Robotics Club of Brac University, Director, Human Resource Department — Apr. 2022 to Mar. 2023

Bangladesh Junior Science Olympiad, Organizer — Dec. 2022 - Mar. 2023

Brac University Computer Club, Assistant Director, Creative Department — Jan. 2022 - Dec 2022

Bangladesh Mathematical Olympiad, Volunteer & Organizer — Jan. 2020 - Dec 2021

Dhaka City College Photography Club, Founder President — Jul. 2017 - Aug. 2018

Dhaka City College Science Club, Head of Photography — Jul. 2017 - Aug. 2018

Ideal Science and Technology Aiming Research Council-ISTARC, Panel Member — May. 2015 - Apr. 2016

Recreational Activities

Street Photography, Mountain Hiking, Cycling, Playing Piano, Listening to Music, Visiting Art & Photography Exhibitions, Walking and explore the world

Professional Courses

BASIS & Skills for Employment Investment Program (SEIP): Mobile Application Development Course using Flutter — Oct. 2021 - Apr. 2022

British Council: Senior Young Learners, Upper-Intermediate (B2) Level Course — Apr. 2016 - May 2016

Workshops/Bootcamp

BracU Programming Skill Bootcamp in Brac University — Summer'2019

Machine Learning using Python by Gyanjam — Feb. 2019

Graphic Design by Creative IT Institute — Mar. 2019

Photography Workshop by Pixmama — Nov. 2018

Summer Science Camp by Society for the Popularization of Science — May 2017

Competitions Attended

Datathon 2.0 by Robi Axiata Ltd.	Dec. 2023
Google Summer of Code	2021 - 2023
Iranian Geometry Olympiad	2019 - 2020
Intra University Programming Contest by Brac University	Spring'2019
Participated in Mathematical Olympiad in National Round	2014 - 2018
Participated in Physics Olympiad in Regional Round	2016 - 2018
Participated in Biology Olympiad in National Round	2017 - 2018
True Festival 1.0 (Photography Competition) by Viqarunnisa School	Mar. 2015

Online Courses

Human Computer Interaction - Theory & Principals (View Certificate)	Udemy
Human-Centered Design: an Introduction (View Certificate)	Coursera
Introduction to Augmented Reality and ARCore (View Certificate)	Coursera
Research Ethics and Compliance Training (View Certificate)	Western Washington University